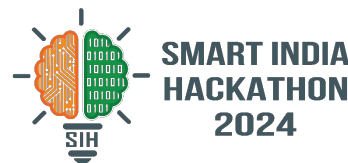
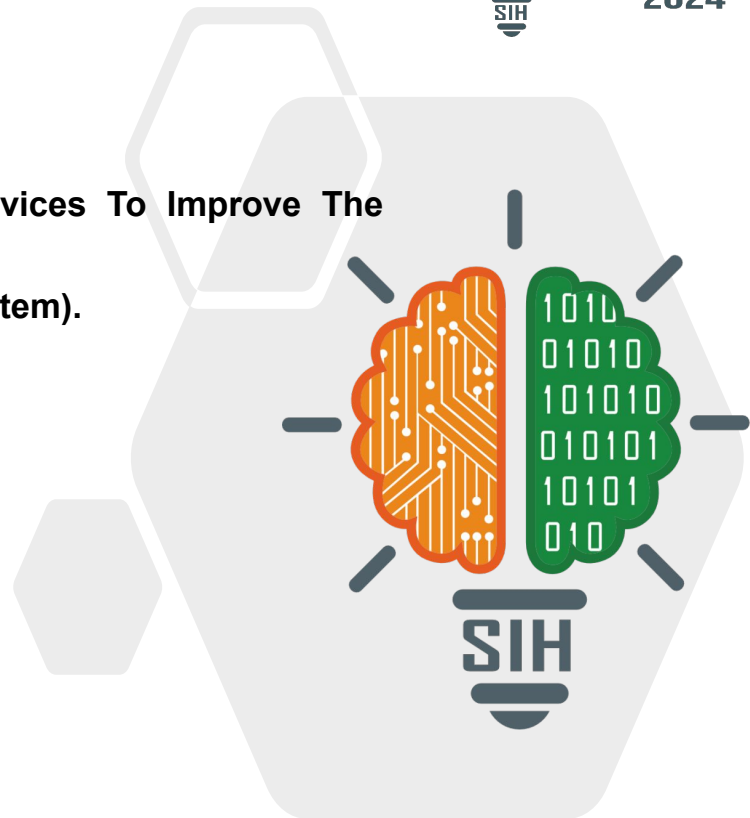


SMART INDIA HACKATHON 2024



- ❑ **Problem Statement ID** –SIH 1535
- ❑ **Problem Statement Title**- Creating Intelligent Devices To Improve The Commutation Sector(Smart AI Based Accident Detection System).
- ❑ **Theme**- Smart Vehicles
- ❑ **PS Category**- Hardware
- ❑ **Team ID**- 4934
- ❑ **Team Name**- AARAMBH



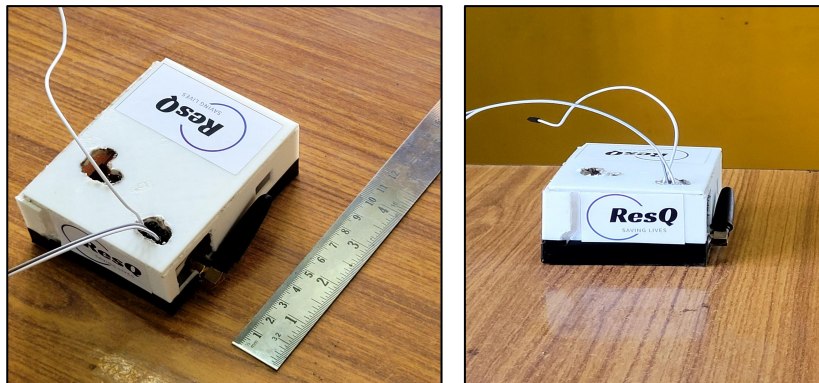
‘ResQ’-Instant Alerts, Instant Safety



Customer Pain Points:

1. **Limited Options for Two-Wheelers:** Existing models focus on four-wheelers, leaving two-wheel riders underserved.
2. **Incomplete Solutions:** Many devices offer only partial functionalities, neglecting crucial features like camera capabilities.
3. **High Cost:** Existing solutions are expensive, limiting access for many two-wheeler riders.
4. **Safety Concerns:** Two-wheelers lack safety features found in cars, and helmet usage is low in our country.
5. **Desire for Comprehensive Protection:** Riders seek solutions that cover accident detection, GPS tracking, and real-time image capture.

FINAL WORKING PROTOTYPE



Our Proposed Solution: 'ResQ' By AARAMBH

- ❑ **Swift Accident Detection:** The system detects accidents using the ADXL335 accelerometer, monitoring changes in the x, y, and z axes to identify crashes or impacts.
- ❑ **Immediate Alerts:** Upon detecting an accident, the Arduino sends an SMS alert via the SIM900A module to predefined contacts such as family members, hospitals, and police stations.
- ❑ **Location Sharing:** The alert includes a Google Maps link generated from the NEO-m8n GPS module, allowing precise identification of the accident location.
- ❑ **Camera Integration:** The ESP32-CAM captures images before, during, and after the accident, documenting the event for analysis.
- ❑ **AI Model for Accident Cause Analysis:** Images captured by the ESP32-CAM are processed using a YOLO-based AI model to determine the cause of the accident (e.g., collision with another vehicle or object). This analysis can also be used for generating insights and reports.
- ❑ **Vehicle Tracking:** The app includes real-time vehicle tracking, accident alerts, and a map feature displaying accident hotspots.
- ❑ **False Alarm Safety Mechanism:** A 30-second buzzer allows the rider to cancel false alarms by pressing a handlebar button, preventing unnecessary location sharing.

TECHNOLOGY STACK:

Microcontroller: Arduino Uno

Accelerometer: ADXL335

- Monitors changes in x, y, and z axes for detecting crash/impact.

GPS Module: NEO-M8N

- Provides precise location data.

Communication Module: SIM900A

- Sends SMS alerts with accident details to predefined contacts.

Camera Module: ESP32-CAM

- Captures pre, during, and post-accident images.
- Images are used for **AI-based analysis (YOLO)** and shared with insurance companies.

Software Development:

- Flutter for cross platform mobile application development.
- Creating communication protocols and a user interface.

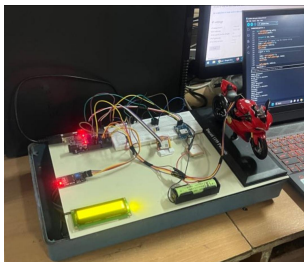
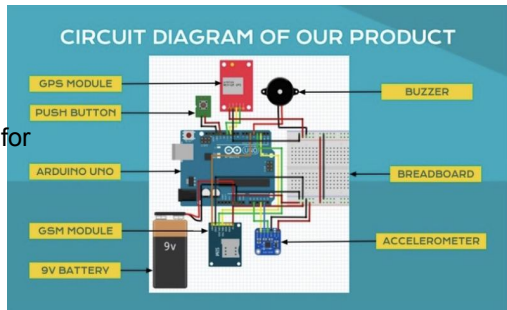
Location Services: Google Maps

- Pinpoints accident locations with shareable links.

Database: NoSQL (Real-time database)

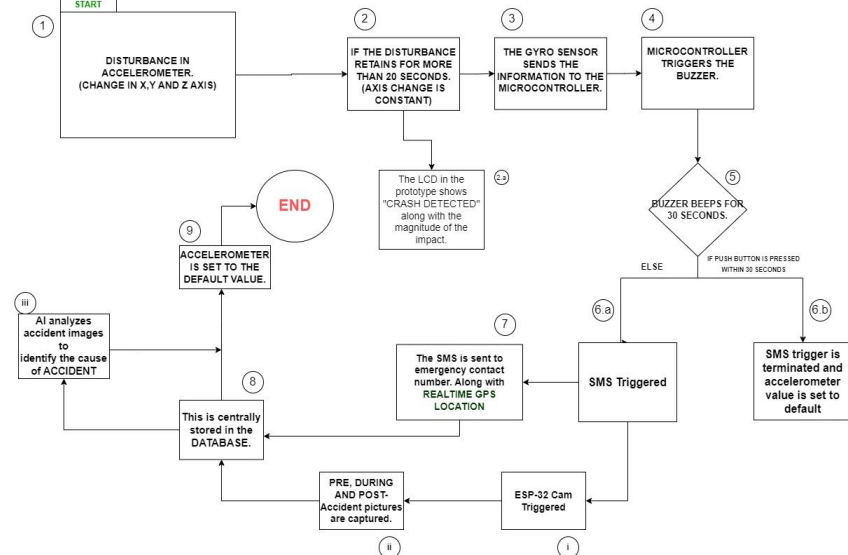
- Centrally stores and accesses data.

Programming Language: C++ , embedded C, Flutter (dart)



INITIAL WORKING PROTOTYPE

ACCIDENT DETECTION PROCESS FLOW.



Politically Acceptable

Aligns with stakeholder interests



Socially Desirable

Benefits community & society



Technologically Feasible

Achievable with current technology



Financially Viable

Sustainable cost-benefit ratio



Administratively Doable

Manageable implementation process



Judicially Tenable

Complies with legal framework

INNOVATION: USP's AND IMAGES OF APP AND DASHBOARD



Instant Accident Detection

Advanced AI-powered sensors detect accidents in real-time, triggering immediate emergency response within seconds.

70% Faster Response Time



Plug & Play Installation

Simple 10-minute installation process with universal compatibility for all two-wheelers. No technical expertise required.

Universal Compatibility



Driver Behavior Analysis

Real-time monitoring of driving patterns, providing insights and alerts for safer riding habits.

Preventive Safety Measures



Accident Prone Areas

Maps and alerts for high-risk zones based on historical data and real-time conditions.

Risk Prevention



AI Camera Integration

Smart camera system captures crucial pre/post accident data for analysis and documentation.

Visual Documentation



Smart Command Center

Unified dashboard for emergency services with real-time tracking and response coordination.

Centralized Control



Live Location Tracking

Real-time GPS tracking with emergency route optimization and location sharing.

Precise Localization



Analytics Dashboard

Comprehensive data analytics for behavior patterns, accident analysis, and safety improvements.

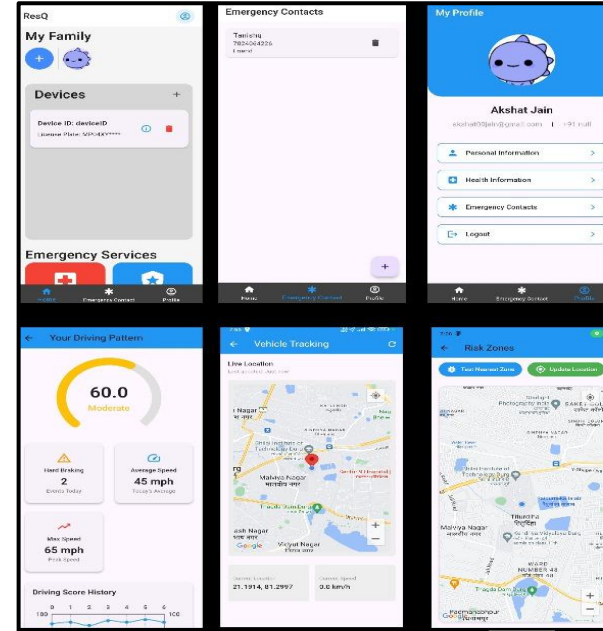
Data-Driven Insights



Emergency Integration

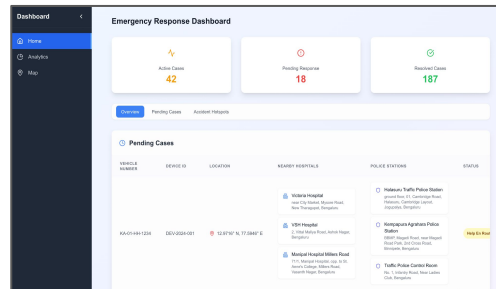
Direct connection with emergency services, hospitals, and family contacts for immediate response.

Rapid Response Network



State of Development/TRL-

- Conceptualization and design phases completed.
- First prototype of the emergency response system for two-wheelers is ready.
- Prototype currently undergoing testing to identify flaws.
- Team actively working on improvements for first prototype iteration.



MARKET ANALYSIS

With over 27 crore two-wheelers in India and rising accident rates, ResQ targets a significant market opportunity in vehicle safety solutions.

TAM (Total Addressable Market)

Considering 20% of total two-wheeler market as potential early adopters

$$17,00,00,000 \times 20\% = 3.4 \text{ Crore Units}$$

SAM (Service Addressable Market)

Accounting for market accessibility and initial reach limitations

$$3.4 \text{ Crore} \times 90\% = 3.06 \text{ Crore Units}$$

Target Market Value

At competitive price point of ₹2,500 per unit

$$₹2,500 \times 3,06,00,000$$

= ₹7,650 Crore

Represents significant market opportunity

BUSINESS MODEL

TARGET USERS:

Primary Users

- Urban Professionals (25-35 years)
- Daily Two-Wheeler Commuters
- Delivery Fleet Riders
- Safety-Conscious Two-Wheeler Owners
- Corporate Transport Users
- Rental Vehicle Users

Business Partners

- Two-Wheeler Manufacturers (OEMs)
- Insurance Companies
- Fleet Management Companies
- Emergency Service Providers
- Smart City Authorities
- Road Safety Organizations

Product Sale

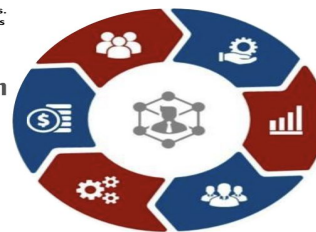
Partnerships with two-wheeler manufacturers (OEMs) to integrate our system into new models. Generate revenue through licensing agreements or per-unit sales to OEMs.

Mobile App & Subscription

Develop a companion mobile app for real-time alerts and ride analytics. Offer subscription-based services within the app for ongoing monitoring and updates.

OEM Integration

Provide subscription-based services for ongoing monitoring, maintenance, and updates. Offer different tiers with varied features and support levels to suit diverse customer needs.



Government Collaborations

Partner with government agencies and road safety organizations to promote adoption. Explore subsidies or incentives to encourage widespread implementation among two-wheeler riders.

Data Analytics

Collect and analyze data from deployed systems. Provide actionable insights to customers, including riders, OEMs, and insurance companies based on accident trends and patterns.

Insurance Partnerships

Collaborate with insurance companies to offer incentives for riders with our system installed. Establish revenue-sharing agreements based on accident data and risk reduction.

COMPANY NAME	ResQ	ROADSATHI	BOSCH	JIO	MAHINDRA AND MAHINDRA	BLACKBOX GPS
PRODUCT	IOT DEVICE	RS SAFETY TAG	DRIVE DROWSIN ALERT SYSTEM	JIO MOTIVE	IN-BUILT	AIS 140 IRNSS DEVICE
PRICE	RS 2500/-	RS 599/-	RS 35000/-	RS 5499/-		RS 13500/-
TARGET AUDIENCE	BIKES CARS	BIKES CARS	CAR	CAR	CAR	ALL VEHICLES
FEATURES	ACCIDENT ALERT CONTACTING EMERGENCY SERVICES CAMERA WITH INTEGRATED AI MODEL USER FRIENDLY APP DRIVER BEHAVIOUR ANALYSIS ACCIDENT PRONE AREA LIVE VEHICLE TRACKING CENTRAL POLICE / AMBULANCE DASHBOARD	WRONG PARKING ALERT UNLOCKED VEHICLE ALERT ACCIDENT ALERT EMERGENCY ALERT	STEERING DRIVER ANALYSIS DRIVER ALERT TIME BAED MONITOING VEHICLE SPEED AND LANE POSITION	HEALTH STATUS MONITORNG ANTI-THEFT GEOFENCING 4G GPS TRACKING ACCIDENT DETECTION	CRASH SENSOR EMERGENCY BUTTON LOCATION SHARING CONTACTING EMERGENCY SEVICES EMERGENCY SENSORS	IRNSS MODULE GEOFENCING GEOTAGGING AUTOMATIC ACCIDENT CALL VEHICLE TRACKING

Component Costs (10,000+ Units)

S.No	COMPONENTS	RETAIL PRICE	BULK PRICE
1	PCB Manufacturing	Rs 150/-	Rs 45/-
2	ATmega328P Microcontroller	Rs 180/-	Rs 85/-
3	SIM900A Module	Rs 900/-	Rs 350/-
4	NEO-M8N GPS Module	Rs 900/-	Rs 400/-
5	ADXL335 Accelerometer	Rs 300/-	Rs 120/-
6	ESP32-CAM Module	Rs 400/-	Rs 180/-
7	Enclosure & Hardware	Rs 150/-	Rs 50/-
SUBTOTAL (A)		Rs 2,980/-	Rs 1,230/-

TOTAL PRODUCTION COST

Components (A)	Rs 1,230/-
Manufacturing (B)	Rs 120/-
Compliance (C)	Rs 70/-
FINAL TOTAL	Rs 1,420/-

From Prototype (Rs 3,000/-) to Production (10,000 U):
Cutting costs to just Rs 1,420/- per unit!

Selling Price: Rs 2,500/-

Gross Margin: ~43%

All prices include GST and are sourced from authorized distributors



Future Scopes / Roadmap



SMART INDIA
HACKATHON
2024

RESQ: 3-YEAR STRATEGIC ROADMAP

2025

Q1-Q2: Foundation

- Patent Filing: Hardware System
- ARAI Certification Initiation
- Beta Testing (100 units)

Milestone: Hardware Patent Application

Q3-Q4: Development

- Algorithm Patent Filing
- Manufacturing Setup
- Startup India Registration
- Initial Grant Applications

Target: Production Ready Status

2026

Q1-Q2: Market Entry

- Production Scale-up
- OEM Partnerships
- Regional Market Launch
- Insurance Collaborations

Goal: 10,000 Units Deployment

Q3-Q4: Enhancement

- AI Model Optimization
- Feature Expansion
- Smart City Integration
- Data Analytics Platform

Target: Enhanced AI System Patent

2027

Q1-Q2: Scaling

- Pan-India Expansion
- R&D Center Launch
- Advanced Features Release
- Government Partnerships

Goal: 100,000 Units Milestone

Q3-Q4: Innovation

- Predictive Analytics Launch
- International Market Prep
- Advanced IoT Integration
- New Patent Applications

Vision: Market Leadership

IMPACT

Saves Lives

Prevents Accidents Using AI

Improves Overall Safety

Enables False Claim Analytics

Eliminates Delays in Emergency Response

Provides Legal & Insurance Support

Grant Program	Amount	Focus Area
Startup India Seed Fund	Up to ₹20 Lakhs	Hardware Startups
NITI Aayog - AIM	Up to ₹10 Lakhs	Innovation & Product Development
IIC YUKTI 2.0	₹5-15 Lakhs	Academic Innovation
DST NIDHI Program	Up to ₹1 Crore	Technology Commercialization
MeitY TIDE 2.0	Up to ₹7 Lakhs	IoT & Hardware
AICTE KAPILA	Up to ₹10 Lakhs	Patent Filing Support
Smart City Innovation	Up to ₹25 Lakhs	Urban Safety Solutions

REFERENCES LINK :

Product explanation video link :
<https://app.animaker.com/animo/GUAhx0ZOIHdeaOAz/>

Impact analysis judgement research paper:
https://drive.google.com/file/d/17DY4r-2ktVG_Jx79TFM1XA-JQGqDM_aB/view?usp=sharing

Accidents records official :
<https://morth.nic.in/road-accident-in-india>

MEET OUR TEAM-AARAMBH



KSHITIJ SINGH



AKSHAT JAIN



TANISHQ SOOD



SHREYASH GOP



VANSHIKA GARG



SHRIYANSHI SONI



(Mentor)

SHIV SINGH BHADOURIYA